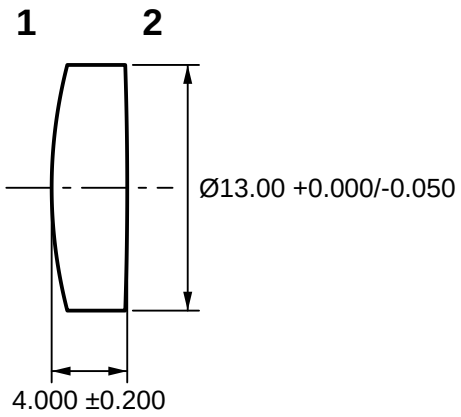

Surface 1 sag			
h	**z**	**±dz µm**	**±ds mrad**
0.00	0.0000	0.00	0.000
1.00	0.0195	0.43	0.866
2.00	0.0780	1.72	1.708
3.00	0.1757	3.83	2.501
4.00	0.3126	6.70	3.221
5.00	0.4889	10.24	4.186
6.00	0.7048	14.35	5.402

NOTES:

1. INTERPRET DRAWING PER ASME Y14.18 AND ASME Y14.5. DIMENSIONS IN MILLIMETERS.
2. MATERIAL: N-BK7 (SCHOTT) OR EQUIVALENT, nd = 1.51680 ± 0.0010, vd = 64.17 ± 0.8%.
3. POWER AND IRREGULARITY IN FRINGES AT 632.8 NM OVER THE CLEAR APERTURE.
4. SURFACE QUALITY 60-40 PER MIL-PRF-13830B OVER THE CLEAR APERTURE.
5. WEDGE: EDGE THICKNESS DIFFERENCE 0.0257 MAX (BEAM DEVIATION 3.51' MAX). SURFACE TILT AS TABLE, REFERRED TO DATUM AXIS OF THE OUTSIDE DIAMETER.
6. PROTECTIVE BEVEL 0.3-0.6 FACE WIDTH X 45°.
7. COATING: SURFACE 1 NONE; SURFACE 2 NONE.
8. EDGE THICKNESS 3.07 REF.
9. SURFACE 1 ASPHERIC: Z = (H²/R)/(1 + sqrt(1 - (1+K)H²/R²)) + sum A_I H^I, K = -0.550000 ±0.005, A4 = -1.2E-06.
10. SURFACE 1 DECENTER 0.100 MAX TO THE DATUM AXIS.
11. SAG TABLES: Z FROM THE VERTEX ALONG THE AXIS, POSITIVE TO THE RIGHT. ±dZ AND ±dS ARE THE MOST THE SURFACE'S POWER AND IRREGULARITY TOLERANCES MOVE THE SAG AND TILT THE SURFACE AT THAT HEIGHT.
12. SURFACE 1: ROTATIONALLY SYMMETRIC IRREGULARITY 0.5 FR MAX, PART OF THE IRREGULARITY (ISO 10110-5 C).
13. IN ITS MOUNT (ASSEMBLY DRAWING): DECENTER X ±0.2 MM, DECENTER Y ±0.2 MM, TILT X ±0.2°, TILT Y ±0.2°.

SURF	RADIUS	CA Ø	POWER	IRREG	TILT	SURF QUAL
1	25.641 ±0.5 CX ASPH	12.00	-	1.5 FR	4.8'	60-40
2	200.000 ±20 CX	12.00	-	1.5 FR	4.8'	60-40

TITLE	**LENS 1-2**
DRAWING	OPT-01
MATERIAL	N-BK7
SCALE	2.5:1 UNITS mm SHEET 1/1
STANDARD	ASME Y14.18 / MIL-PRF-13830B
PROJECT	aspheric_singlet
DRAWN	2026-10-15